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**ÚSTAV POČÍTAČOVÝCH SYSTÉMŮ**

# 3D NEURAL CELLULAR AUTOMATA

**3D NEURÁLNÍ CELULÁRNÍ AUTOMATY**

**BACHELOR'S THESIS**
**BAKALÁŘSKÁ PRÁCE**

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**BRNO 2026**

## Abstract

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v anglickém jazyce.

## Abstrakt

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v českém (slovenském) jazyce.



## Keywords

Sem budou zapsána jednotlivá klíčová slova v anglickém jazyce, oddělená čárkami.

## Klíčová slova

Sem budou zapsána jednotlivá klíčová slova v českém (slovenském) jazyce, oddělená čárkami.


## Reference

REPČÍK, Michal. *3D Neural Cellular Automata*. Brno, 2026. Bachelor's thesis. Brno University of Technology, Faculty of Information Technology. Supervisor Ing. Karel Fritz,

# 3D Neural Cellular Automata

## Declaration

I hereby declare that this Bachelor's thesis was prepared as an original work by the author under the supervision of Ing. Karel Fritz I have listed all the literary sources, publications and other sources, which were used during the preparation of this thesis. I have used ChatGPT to correct spelling and other language mistakes. I used Grok when working on the software.

. . . . . .
Michal Repčík
November 17, 2025


## Acknowledgements

I would like to express my sincere thanks to my supervisor, Ing. Karel Fritz, for his guidance and support.

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# Theoretical Foundations

## 3.1 Prerequisites

This thesis assumes the reader is familiar with basic concepts of deep learning, in particular convolutional neural networks, gradient-based optimisation, and automatic differentiation in PyTorch [5, 1].

A comprehensive introduction to these topics can be found in Goodfellow et al. [2] and in the official PyTorch documentation [1].

The rest of the thesis is self-contained with respect to cellular automata and their neural extensions.

## 3.2 Cellular Automata

Cellular automata (CA) are discrete computational systems that consist of a regular grid of cells, each of which can take on a finite number of states. The state of each cell evolves in parallel over discrete time steps according to a local update rule, which depends on the states of its neighboring cells [4]. Formally, a cellular automaton is defined by:

- A lattice of cells, typically one-, two-, or three-dimensional.
- A finite set of states $\mathcal{S}$ that each cell can occupy.
- A neighborhood function $\mathcal{N}$ that defines which cells influence a given cell.
- An update rule $f : \mathcal{S}^{|\mathcal{N}|} \to \mathcal{S}$ that computes the next state [3].

# Chapter 4

# Conclusion

TODO

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